



Viewers Had To Look Away When This Happened
Live TV
Stars Are Made

Broadcast a Signal on a Magnetic Field.

A Magnetic field created with an audio AC voltage can be picked up and heard with a pickup coil plugged into a small amplifier. This magnetic field can be made to fill an entire house by running one turn of copper wire around the house and driving it with the speaker output of a small transistor radio.

Why does this magnetic field fill the entire house?

This experiment shows that the size of a magnetic field has nothing to do with its strength but with the size of the coil that produces it. Try the above experiment again but instead of driving the wire around the house, drive a loop that is only 3 inches in diameter and see how far you can get and still hear the signal. You will only get a few feet at best. Imagine having a loop of wire around the earth! The earth's magnetic field is a large one; one that acts as though it were produced by a huge current carrying coil wrapped around the earth out in space. The earth's magnetic field is not a strong one but an observer can travel many miles without noticing much difference in its intensity. A magnetic field produced by a small coil will diminish greatly when observed only a few feet away. If one could make a coil around the earth and drive it as in the above experiment, you could probably pick up the sound from anywhere on the planet.

The drawing shows why this is so. Notice that all of the lines inside of the coil are all running the same direction. These lines will all have a similar influence on a pickup coil that is placed anywhere in the coil. Notice however, that the lines of force at a considerable distance from the coil tend to be going in opposite directions and thus tend to cancel each other.

So now we can start to see why radio waves travel so far!

It is my belief that radio waves are nothing more than a simple magnetic field as described above, that is produced as though it came from part of an infinitely large coil. How can this be. Normally, we think of a complete circuit being required in order to carry current through a wire. It is this current that produces the magnetic field. In the case of AC however, current can flow through a wire back and forth if the wire is long enough. A complete circuit is just plain not necessary. At radio frequencies, the wire does not have to be very long in order to get current to flow back and forth without a complete circuit. A simple piece of wire stuck into the air will radiate a magnetic field and thus radiate a radio signal if it is driven with an ac signal that is high enough in frequency to make significant current flow through it. There is no "other side" of the coil to create any cancelling magnetic fields as you get far away from the antenna.

Rite about now you may be asking, "What about a Loop Antenna?" The answer is that the transmitting Loop Antenna is usually of significant size in comparison to the wavelength of the radio signal being radiated. This means that the magnetic fields from the loop cancel in some places and actually reinforce each other in other places.

That's it!! A piece of wire stuck up in the air makes a great antenna as long as an ac current can be driven through it. The only reason very low frequencies are not normally used for radio communication, is because of the prohibitively long lengths of wire required to make an antenna. If the wire is not long enough, the current arrives at the end long before it is time to reverse and flow in the opposite direction. At high frequencies, the current keeps reversing before the electrons can travel very far in the wire and thus AC current flow is possible in a reasonably short wire.

Misconception No. One!!

1. Radio frequencies are always higher than those of sound.

When we are taught about the spectrum of frequencies, we tend to get the misconception that radio frequencies are always higher than audio frequencies and that the main difference between an audio signal and a radio signal is their frequencies. Nothing can be farther from the truth. The main difference between an audio signal and a radio signal is that an audio signal is vibrating air and a radio signal is a vibrating magnetic field. Radio frequencies are part of the electromagnetic spectrum and sound is part of the acoustic spectrum. Even though Sound and Radio signals involve two separate mediums or spectra, their frequency ranges can certainly overlap. Audio signals in the air can be produced at frequencies well into the multi tens of kilocycles, while radio signals can be produced at frequencies way down into the audio spectrum or lower. As stated above, radio signals in the audio frequency range are less common because of the very long antennas needed to radiate them.

Misconception no. Two!!

Standing Wave Ratio is a sacred antenna parameter!!!

If you study a lot of antenna literature, particularly that written for Ham Radio operators, you will get the impression that proper Standing Wave Ratio is the sacred key to good antenna performance. Actually, Standing Wave meters are mostly the result of modern radio transmitters being designed to drive only 50 ohm output loads. The easiest way to get a finger on matching an antenna system to this 50 ohm transmitter is to use a meter that tells if the load is matched to 50 ohms. Someone decided to get everyone involved in looking at the concept of "Standing Wave Ratio" instead of that of "Impedance Matching". In a way, the SWR meter gives an indication of both. It would have been a lot easier if they had put the emphasis on the concept of "Impedance matching" instead of "Standing Wave Ratio". The impedance matching concept is much more intuitive and is more of the real issue than the concept of standing wave ratios. The standing wave concept, being less intuitive, does not get to the real issue at hand. Many "Experts" talk above everyone and sound smart because no one can understand them. Of course they are hard to understand because they are not always making complete sense.

Impedance matching, by the way, is basically a concept of how some antennas can act like a 12 volt 50 watt light bulb or a 110 volt 50 watt light bulb. A power source for the 12 volt bulb will need to supply more current at less voltage and the power supply for the 110 volt bulb will need to supply more voltage at less current. Since different antennas can have different impedances, the transmitter must be able to supply power at different impedances.

This misconception about standing wave ratio is not just my own opinion. A good book on the subject is written by Walter Maxwell W2DU, an antenna specialist in the space program. His book is called "Reflections" and is published by the American Radio Relay League. It goes into great detail about the misconceptions surrounding "Standing Wave Ratio".

One of the best sources of Ham Radio literature on antennas is the Fifteenth edition of the ARRL Antenna Book copyright 1988. This edition finally makes a lot more sense when it comes to antennas.

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